

LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

SAUERS

We give a negative answer to the longstanding question whether every countable group is MF. Here MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. Our main tool is the following one-sided compression criterion. Let G be countable, let $L \leq G$ have property (T), and let $K \trianglelefteq G$ have property (T). If K is contained in the normal subgroup generated by the commutators $[ucu^{-1}, \ell]$, where $uLu^{-1} \leq L$, $c \in C_G(L)$, and $\ell \in L$, then every homomorphism from G to an MF group kills K . For $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$, a single commutator of this form normally generates H . The group H is nontrivial, simple, and has property (T), whereas every homomorphism from H to an MF group is trivial. In particular, H is not MF. Consequently, the reduced group C^* -algebra $C_r^*(H)$ is separable and stably finite, but is not MF. Finally, if $f: G \rightarrow Q$ is surjective and $\ker f \leq \text{Rad}_{\text{MF}}(G)$, then $\text{Rad}_{\text{MF}}(G) = f^{-1}(\text{Rad}_{\text{MF}}(Q))$. In this case, image and inverse image identify the normal subgroups $N \trianglelefteq G$ satisfying $\ker f \leq N$ and G/N MF with the normal subgroups $P \trianglelefteq Q$ satisfying Q/P MF.

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1. INTRODUCTION

Not every countable group is MF. Our counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^* -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is MF if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G),$$

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and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

Thus the models become multiplicative in operator norm without losing any nonidentity element. This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the c_0 -direct sum.

A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona. For every countable group G , the reduced group algebra $C_r^*(G)$ is separable and has a faithful canonical trace, hence is stably finite. If $C_r^*(G)$ is MF, restricting an MF embedding to its canonical group unitaries shows that G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^* -algebra that is not MF.

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

Proposition 1.1 shows that G is MF precisely when $\text{Rad}_{\text{MF}}(G)$ is trivial. If $\text{Rad}_{\text{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF kernel closure* by

$$\text{cl}_{\text{MF}}^G(N) = \bigcap \{\ker f : N \leq \ker f, f: G \rightarrow M, M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G) = \text{cl}_{\text{MF}}^G(1)$, and G/N is MF precisely when $\text{cl}_{\text{MF}}^G(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 1.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \text{cl}_{\text{MF}}^G(N) = N \quad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G , $x \in \text{Rad}_{\text{MF}}(G)$, and π is a corona homomorphism of G , then $\pi \circ \alpha$ kills x . Hence $\pi(\alpha(x)) = 1$ for every π , which proves full invariance.

Put $R = \text{Rad}_{\text{MF}}(G)$ and enumerate the nonidentity elements of G/R as $(x_j)_{j \geq 1}$ and the pairs in $(G/R)^2$. For each x_j , the definition of R supplies

a corona homomorphism whose value at x_j has positive distance from the identity. Every such homomorphism kills R and therefore descends to G/R . Choose coordinate unitary lifts, using polar correction as in Lemma 4.1. At stage n , take a direct sum of one coordinate from each of the first n models. Choose each coordinate far enough out that the first n multiplication defects are at most $1/n$ and the designated value at x_j retains at least half of its corona norm separation. Such coordinates exist arbitrarily far out: the defects converge to zero, while the quotient norm is a limsup. The resulting operator norm asymptotic representation detects every x_j , so its corona homomorphism is faithful on G/R . Thus G/R is MF.

Applied to G/N , the same argument says that G/N is MF exactly when the intersection of the kernels of all maps into MF groups that kill N is N , which is the asserted closure criterion. $\square$

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

We sketch the argument. Corollary 3.4 gives $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$. Equivalently, if (V_n) is any operator norm asymptotic representation of G , then $\|V_n(d) - 1\|_2 \rightarrow 0$ for every $d \in \mathfrak{D}_G(L)$. To upgrade this conclusion to corona triviality on K , fix a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ and suppose, toward a contradiction, that Θ is nontrivial on K . Let p be the image in $\mathcal{Q}_{\mathbf{d}}$ of the Kazhdan projection of K . Because $K \trianglelefteq G$, the projection p commutes with $\Theta(G)$; hence $q = 1 - p$ defines a G -invariant complementary corner. Lemma 4.1 realizes the action on this corner by an operator norm asymptotic representation (W_n) on matrix corners with identities q_n . The corner has no nonzero K -fixed vectors, so the Kazhdan inequality yields a single $s_0 \in K$ and a subsequence along which $\|W_n(s_0) - q_n\|_2$ stays bounded below by a positive constant. This contradicts $s_0 \in K \leq \mathfrak{D}_G(L)$ and the preceding convergence. Thus every corona homomorphism kills K , as required.

Theorem 2.1 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the binary Leavitt construction that produces the group H announced above.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^ -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.*

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \text{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the elementary commutator relations show directly that d normally generates H . Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\text{Rad}_{\text{MF}}(H) = H$.

If $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \text{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 3.3.

A full MF radical also lets us prescribe the quotient visible to MF.

Theorem C (prescribed quotients visible to MF). *Let B be a countable group with $\text{Rad}_{\text{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B . Put $A = \langle d \rangle$ and, for a countable group Q , define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q: W_Q \rightarrow Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M .

Relation to prior work. The operator-algebraic lineage of the problem predates the term *MF group*. In a 1994 Oberwolfach abstract, Blackadar and Kirchberg suggested that every separable nuclear stably finite C^* -algebra might be NF [BK94]. Their 1997 paper introduced MF C^* -algebras; for separable nuclear algebras, the MF and NF properties are equivalent [BK]. They also observed that no separable stably finite C^* -algebra was known not to be MF. The broader question whether every separable stably finite C^* -algebra is MF became known as the Blackadar–Kirchberg MF problem [RS].

Carrión–Dadarlat–Eckhardt later introduced the precise group property used here [CDE]. Subsequent approximation literature attributed the question whether every countable group is MF to Kirchberg [LO, Th]. The negative solution of the Connes embedding problem also produced stably finite non-MF C^* -algebras in general [JNVWY]. Theorem B provides such an example among reduced group C^* -algebras and answers the operator norm group approximation question negatively. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves MF permanence for doubles $G *_H G$ of MF groups. Separately, Shulman proves MF full group C^* -algebra results for several classes of amalgamated and semidirect products [Sh]. The operator norm case remained open after failure of approximation had been established for finite Schatten norms [DGLT, LO, Th]. Bachner–Dogon–Lubotzky study stability between the operator and Hilbert–Schmidt norms as a possible source of non-MF groups [BDL]. Here operator norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. The Kazhdan projection of this conjugation representation makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Theorem 4.2 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser’s work gives general theorems about normal subgroups of linear groups over exchange rings [Pre]; the simplicity proof below instead extracts an elementary transvection directly from a nonidentity normal element. Ershov–Jaikin–Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1].

Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. ONE-SIDED COMPRESSION IN FINITE DIMENSION

Theorem 2.1 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. $\square$

The finite-dimensional hypothesis is essential here. On an infinite-dimensional space an injective endomorphism of the commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity for projections onto fixed vectors.

3. KAZHDAN TRANSPORT IN NORMALIZED HILBERT–SCHMIDT NORM

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 3.1. *For every sequence (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A . Consequently $ww^* = q$, and hence $p = q$. $\square$

Lemma 3.2 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 3.3 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the operator norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0 . The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . The preceding estimate shows directly that $g \mapsto \tilde{\sigma}(g)$ is an exact homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$, so U commutes with P . Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 3.4. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 3.3, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

4. NORMAL KAZHDAN SUBGROUPS AND THE MF RADICAL

Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona representation of G . Normality of K implies that p commutes with the image of G . Hence $q = 1 - p$ also commutes with that image, and the corner $q\mathcal{Q}_{\mathbf{d}}q$ has no nonzero K -fixed vectors.

Lemma 4.1 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator norm asymptotic representation*

$$W_n: G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator norm asymptotic representation. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates. $\square$

Theorem 4.2 (normal Kazhdan radical theorem). *Let G be countable, let $D \leq R_{\infty \rightarrow 2}(G)$, and let $K \trianglelefteq G$ have property (T) with $K \leq D$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g) \text{Fix } \Theta(K)$ and $\text{Fix } \Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 4.1 gives an operator norm asymptotic representation (W_n) on nonzero corners $q_n M_{d_n}(\mathbb{C}) q_n$.

Choose a finite symmetric Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. In the corner $qQ_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (q\Theta(s)q - q)^*(q\Theta(s)q - q)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|} q.$$

Indeed, in every representation of the corner there are no K -fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - q_n)^*(W_n(s) - q_n)$$

represent b . Taking the normalized trace on $q_n M_{d_n}(\mathbb{C}) q_n$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - q_n\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)q_n$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0 \in K \leq D \leq R_{\infty \rightarrow 2}(G)$. Thus every corona homomorphism kills K , as required. $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 3.4 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of $R_{\infty \rightarrow 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Theorem 4.2 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. $\square$

Proposition 4.3 (functoriality and normal generation). *Let $L \leq G$ and put $D = \mathfrak{D}_G(L)$.*

If $f: G \rightarrow Q$ is a homomorphism, then

$$(4) \quad f(D) \leq \mathfrak{D}_{f(G)}(f(L)).$$

If $S \leq G$ is a nontrivial simple subgroup and $D \cap S \neq 1$, then $S \leq D$. If in addition the normal closure of S in G is all of G , then $D = G$. If moreover G is countable and both L and G have property (T), then

$$\text{Rad}_{\text{MF}}(G) = G.$$

Alternatively, suppose that f is onto, $S \leq D$ is simple, $f(S) \neq 1$, and $f(S)$ normally generates Q . Then

$$\mathfrak{D}_Q(f(L)) = Q.$$

If, in addition, G is countable and both L and G have property (T), then

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Proof. If $u \in \text{Comp}_G(L)$, then $f(u)f(L)f(u)^{-1} \leq f(L)$. If $c \in C_G(L)$, then $f(c) \in C_{f(G)}(f(L))$. Therefore f takes every defining generator of D into a defining generator of $\mathfrak{D}_{f(G)}(f(L))$. This proves (4).

The subgroup D is normal in G , so $D \cap S$ is normal in S . Simplicity and $D \cap S \neq 1$ give $D \cap S = S$, hence $S \leq D$. Since D is normal in G , it then contains the normal closure of S . When the normal closure of S is G , this proves $D = G$. Under the property (T) hypotheses, Theorem A with $K = G$ then gives $\text{Rad}_{\text{MF}}(G) = G$.

For the image statement, (4) gives $f(S) \leq \mathfrak{D}_Q(f(L))$. The latter subgroup is normal in Q , so it contains the normal closure of $f(S)$ and is therefore all of Q . Property (T) passes to quotients, so the final hypotheses give property (T) for both $f(L)$ and Q . Applying Theorem A inside Q , with $K = Q$, proves the last assertion. $\square$

5. THE BINARY LEAVITT SELF-COMPRESSION

Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (2) give

$$(5) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\text{GL}_3(R)$ with the subgroup of $\text{GL}_{12}(R)$ consisting of the matrices $\text{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$(6) \quad \Psi(A) = qI_3 + s_0 A t_0.$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q s_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = q I_3 + s_0 I_3 t_0 = q I_3 + p I_3 = (p + q) I_3 = I_3,$$

so Ψ is unital. Finally, $t_0 \Psi(A) s_0 = A$, and hence Ψ is injective.

To implement Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$(7) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(8) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

Lemma 5.1. *The matrix τ defined in (7) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. Put $I = I_6$. The Whitehead factorization gives

$$(9) \quad \mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^5 e_{r,6+s}(b_{rs}), \quad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^5 e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (9) therefore belongs to $\mathrm{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is τ , and the result follows. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 5.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(10) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 5.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (8) and (6) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (10) follows. $\square$

6. SIMPLICITY AND THE FULL MF RADICAL

We first prove that H is simple. The argument uses the following consequence of pure infiniteness. For every $0 \neq x \in R$, there are $a, b \in R$ such that

$$(11) \quad axb = 1.$$

Indeed, R is purely infinite simple, so this follows from [AA, Proposition 10(v)] by taking the target element to be 1.

Lemma 6.1 (Normal generation by an elementary transvection). *Let S be a unital ring, let $n \geq 3$, and let $N \trianglelefteq \text{EL}_n(S)$. Suppose that $e_{ij}(x) \in N$ and that $axb = 1$ for some $a, b \in S$. Then $N = \text{EL}_n(S)$.*

Proof. Choose k different from i and j . Normality of N and the elementary commutator relations give

$$[e_{ij}(x), e_{jk}(b)] = e_{ik}(xb), \quad [e_{ji}(a), e_{ik}(xb)] = e_{jk}(1),$$

so $e_{jk}(1) \in N$.

Products of the generalized permutation matrices

$$w_{uv} = e_{uv}(1)e_{vu}(-1)e_{uv}(1)$$

conjugate $e_{jk}(1)$ to $e_{uv}(1)$ or its inverse for any $u \neq v$. Normality of N , and taking inverses if necessary, therefore gives $e_{uv}(1) \in N$ for every $u \neq v$. Given $r \in S$ and $u \neq v$, choose h different from u and v . Then

$$[e_{uh}(r), e_{hv}(1)] = e_{uv}(r) \in N.$$

Thus N contains every elementary generator of $\text{EL}_n(S)$. $\square$

Lemma 6.2 (Coefficient separation). *If $r, s \in R$ are nonzero, then there exist $a, b \in R$ such that*

$$(12) \quad arb = 0, \quad bsar \neq 0.$$

Proof. By (11), choose $u, v, c, d \in R$ such that

$$urv = 1, \quad csd = 1,$$

and set

$$a = dt_1u, \quad b = vs_0c.$$

The Leavitt relation $t_1s_0 = 0$ gives

$$arb = dt_1(urv)s_0c = dt_1s_0c = 0.$$

On the other hand,

$$bsar = vs_0(csd)t_1ur = vs_0t_1ur.$$

If this element were zero, then

$$0 = (ur)(vs_0t_1ur)v = (urv)s_0t_1(urv) = s_0t_1,$$

contrary to $t_0(s_0t_1)s_1 = 1$. $\square$

Proposition 6.3. *The group $H = \text{EL}_{12}(R)$ is nontrivial and simple.*

Proof. Let $N \trianglelefteq H$ be nontrivial. By (11) and Lemma 6.1, it is enough to find $i \neq j$ and $0 \neq x \in R$ such that $e_{ij}(x) \in N$.

The diagonal case. Choose $1 \neq g \in N$ and suppose first that g is diagonal. If g commuted with every $e_{ij}(a)$, then commuting with the $e_{ij}(1)$ would force $g = \lambda I_{12}$. Commuting with arbitrary $e_{ij}(a)$ would then make λ a central unit. But $Z(R) = \mathbb{F}_2$ by [AC, Corollary 4.3], so $\lambda \in \mathbb{F}_2^\times = \{1\}$ and $g = 1$, a contradiction. Thus for some $i \neq j$ and $a \in R$, conjugation by g sends $e_{ij}(a)$ to $e_{ij}(y)$ with $y \neq a$. Consequently

$$[g, e_{ij}(a)] = e_{ij}(y - a) \in N$$

is the required elementary transvection.

The nondiagonal case. Suppose now that g is not diagonal, and choose $\ell \neq i$ with $g_{\ell i} \neq 0$. Put $r = (g^{-1})_{\ell i}$.

If $r = 0$, set

$$A = gE_{i\ell}g^{-1}, \quad B = E_{i\ell}.$$

Here $A^2 = B^2 = AB = 0$. Expanding the commutator gives

$$(13) \quad [ge_{i\ell}(1)g^{-1}, e_{i\ell}(1)] = 1 - BA$$

and its image in H/N is $[e_{i\ell}(1), e_{i\ell}(1)] = 1$. Thus $1 - BA \in N$. Its defect $-BA$ is square-zero and supported in row i . There is an m for which $g_{\ell i}(g^{-1})_{\ell m} \neq 0$: otherwise the (ℓ, ℓ) entry of $g^{-1}g = 1$ would give $g_{\ell i} = 0$. Thus the (i, m) entry of the defect in (13) is nonzero.

If $r \neq 0$, apply Lemma 6.2 to r and $s = g_{\ell i}$, obtaining a, b with $arb = 0$ and $bsar \neq 0$. Set

$$A = g(aE_{i\ell})g^{-1}, \quad B = bE_{i\ell}.$$

Every column of AB except possibly column ℓ is zero, and

$$(AB)_{k\ell} = g_{ki}arb = 0 \quad (0 \leq k \leq 11).$$

Thus $AB = 0$. Since $A^2 = B^2 = 0$, direct expansion gives

$$(14) \quad [ge_{i\ell}(a)g^{-1}, e_{i\ell}(b)] = 1 - BA.$$

In H/N , the left-hand side is the commutator of $e_{i\ell}(a)$ and $e_{i\ell}(b)$, hence is trivial. Therefore $1 - BA \in N$. The defect is square-zero and supported in row i , while its (i, i) entry is $-bg_{\ell i}a(g^{-1})_{\ell i} = -bsar \neq 0$.

Both nondiagonal cases therefore produce an element $1 + v \in N$ with $v^2 = 0$. The matrix v is supported in a single row i and has some nonzero entry v_{im} . Choose n different from both i and m (with only $n \neq i$ needed when $m = i$). Direct multiplication gives

$$[1 + v, e_{mn}(1)] = e_{in}(v_{im}) \in N.$$

Since $v_{im} \neq 0$, Lemma 6.1 gives $N = H$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. Every elementary generator of the embedded subgroup L has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that each such generator commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (7) gives

$$(15) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since $t_1 q s_1 = 1$, Lemma 6.1, applied to $e_{02}(q)$, gives $\langle\langle d \rangle\rangle_H = H$. $\square$

Proof of Theorem B. The finitely generated $\mathbb{F}_2$ -algebra R is countable, and hence so is H . The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (10), the centrality of c relative to L , and Proposition 6.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 6.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Independently of this radical calculation, Proposition 6.3 shows that H is nontrivial and simple.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful corona embedding of M . Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C_r^*(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If $C_r^*(H)$ were MF, restricting an MF embedding to the canonical unitaries would make H an MF group, a contradiction. $\square$

7. QUOTIENTS VISIBLE TO MF

The MF radical is exact across every surjection whose kernel is already invisible to MF.

Proposition 7.1 (full-kernel pullback). *Let $f: G \rightarrow Q$ be a surjective homomorphism of countable groups. If $\ker f \leq \text{Rad}_{\text{MF}}(G)$, then*

$$\text{Rad}_{\text{MF}}(G) = f^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

In particular, the conclusion holds whenever $\text{Rad}_{\text{MF}}(\ker f) = \ker f$. More generally, for every normal subgroup $N \trianglelefteq G$,

$$\text{cl}_{\text{MF}}^G(N) = f^{-1}(\text{cl}_{\text{MF}}^Q(f(N))).$$

Consequently,

$$G/N \text{ is MF} \iff \ker f \leq N \text{ and } Q/f(N) \text{ is MF.}$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f . Conversely, composing a corona homomorphism of Q with f gives one of G . Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the final assertion, functoriality of the MF radical under the inclusion $\ker f \rightarrow G$ gives $\ker f \leq \text{Rad}_{\text{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N , gives the closure formula. If this closure equals N , then it contains $\ker f$. Mapping the equality onto Q shows that $\text{cl}_{\text{MF}}^Q(f(N)) = f(N)$. Conversely, these two conditions give

$$\text{cl}_{\text{MF}}^G(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 1.1, applied in G and Q . $\square$

Thus image under f and inverse image under f give mutually inverse correspondences, preserving inclusion between the normal subgroups with MF quotient in G and in Q . The map induced by f on the largest quotients visible to MF is an isomorphism:

$$G/\text{Rad}_{\text{MF}}(G) \cong Q/\text{Rad}_{\text{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with kernels invisible to MF.

Let B be a countable group with $\text{Rad}_{\text{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B . Put $A = \langle d \rangle$. For a countable group Q , define

$$(16) \quad W_Q = B *_A (Q \times A),$$

where $A \rightarrow Q \times A$ is the inclusion into the second factor. The trivial map $B \rightarrow Q$ and the projection $Q \times A \rightarrow Q$ agree on A . They therefore define an epimorphism

$$\pi_Q: W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \rightarrow Q \times A \rightarrow W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (16) are injective; in particular, $d \neq 1$ in W_Q .

Proposition 7.2 (universal factorization). *Let T be a group such that every homomorphism $B \rightarrow T$ is trivial. Then precomposition with π_Q is a bijection*

$$\text{Hom}(Q, T) \longrightarrow \text{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f: W_Q \rightarrow T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q . The universal property of the amalgam shows that f factors through π_Q . The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills B , since $\langle\langle d \rangle\rangle_B = B$, and kills the A -factor of $Q \times A$. The resulting quotient is Q , with quotient map induced by π_Q . Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. $\square$

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\text{Rad}_{\text{MF}}(B) = B$. Proposition 7.2 therefore gives the asserted bijection for every MF group M . In addition, the definition of $\text{Rad}_{\text{MF}}(B) = B$ says directly that every homomorphism from B to $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is trivial. Applying Proposition 7.2 with $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ shows that every corona homomorphism from W_Q factors uniquely through π_Q . Intersecting their kernels gives

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d . Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

$\square$

If $N \trianglelefteq W_Q$, the same factorization gives

$$(17) \quad \text{cl}_{\text{MF}}^{W_Q}(N) = \pi_Q^{-1}(\text{cl}_{\text{MF}}^Q(\pi_Q(N))).$$

Indeed, a homomorphism from W_Q to an MF group kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (17) give

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \leq N$, then $N = \pi_Q^{-1}(\pi_Q(N))$.

Taking $Q = \mathbb{Z}$ makes the quotient criterion especially concrete: for every normal subgroup $N \trianglelefteq W_{\mathbb{Z}}$,

$$(18) \quad W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N.$$

Indeed, every quotient of $\mathbb{Z}$ is MF, while $\ker \pi_{\mathbb{Z}} = \langle\langle d \rangle\rangle_{W_{\mathbb{Z}}}$.

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USE OF AI AND FORMAL METHODS

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